

EcoMetrics

Egypt

— *Turning Waste Into Data. Data Into Action.*

A Business Intelligence framework diagnosing Egypt's recycling bottlenecks — from collection trucks to sorting conveyors — built to accelerate Egypt Vision 2030.

100M+
TONS WASTE / YEAR

37%
RECYCLING RATE (2024)

60%
VISION 2030 TARGET



Built within a national digital pioneers initiative.

EcoMetrics Egypt is a Business Intelligence framework for Microsoft Power BI, developed under Egypt's official capstone pathway for data analytics talent.

01

Digital Egypt Pioneers Initiative

DEPI • depi.gov.eg

Institutional framework under the Ministry of Communications and Information Technology (MCIT), Arab Republic of Egypt — a national initiative building a generation of digital pioneers across software, data analytics, AI, and more.

02

Computik Learning Solutions

LEARNING PARTNER

Provided the structured learning environment, technical resources, and curriculum scaffolding for the DEPI Data Analytics capstone program from inception to delivery.

03

Ministry of Communications & IT

MCIT • mcit.gov.eg

The parent ministry overseeing the DEPI initiative nationally — anchoring the project's relevance to Egypt's broader digital transformation strategy.

Five researchers. Two cities. One **dataset**.

The EcoMetrics Egypt project was built by five data analytics researchers from the DEPI Data Analytics Track, spread across Giza and Cairo, under expert supervision.



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GIZA, EGYPT



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Eng. Mahmoud Seraj — Expert Supervisor

Technical guidance and supervision throughout the project lifecycle, under the DEPI program via Computik Learning Solutions.

100 million tons. Every single **year**.

Egypt generates more than 100 million tons of solid waste annually — one of the highest volumes in the MENA region — driven by a population exceeding 100 million and rapid urbanization across Cairo, Giza, and the Nile Delta. Every ton dumped is a ton of recoverable economic value permanently lost.

80–88%

of all waste ends up in open landfills and unregulated dumps across the country.

5M

tons of plastic waste generated per year — a single material stream demanding dedicated sorting capacity.

EGP 1–2B

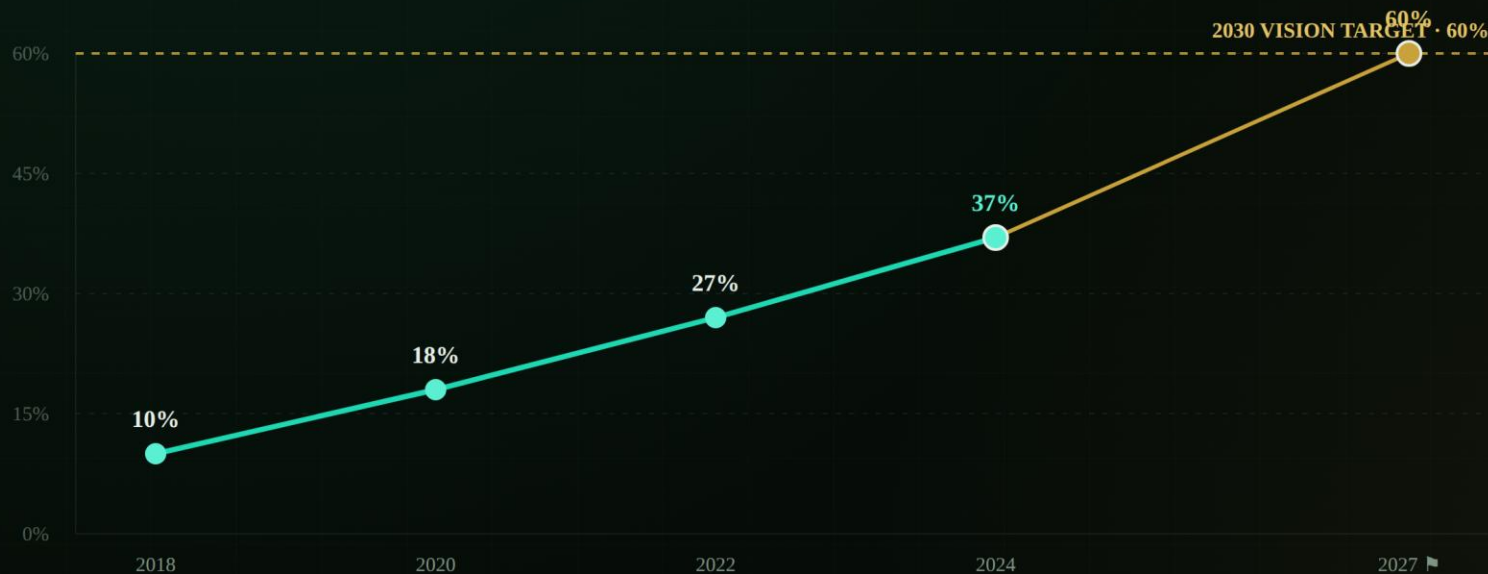
committed by the Egyptian government to recycling infrastructure investment.

100M+

people — Egypt's population, exceeding 100 million, with continuing rapid urban growth.

Six years of progress, plotted in full.

National recycling rate, 2018–2027. From a 10% baseline — where 90% of waste was unsustainably disposed — to a 37% rate in 2024, en route to Egypt Vision 2030's formal target of 60% by 2027.



- 2018
National baseline — 90% of waste unsustainably disposed.
- 2020
Early sorting infrastructure investments begin showing results.
- 2022
Public-private partnerships accelerate sorting capacity.
- 2024
3.7× improvement over 2018 — but 23 points short of target.

37% · 23-POINT GAP · 60% BY 2027

*“The gap between 37% and 60% will not close on its own
—
but data can show exactly where to push.”*

37%

CURRENT (2024)

23 pts

GAP TO CLOSE

60%

TARGET (2027)

Four deliverables. One framework.

The core objective: build a data-driven BI framework that diagnoses recycling bottlenecks from collection to sorting to processing, identifies exactly where value is lost in the pipeline, and quantifies the interventions with the highest impact per unit of investment.

01

Power BI Dashboard

.PBIX • ~1 MB

Three interactive dashboard pages with cross-page filtering, drill-through, and geographic heat maps.

02

Research Paper

.PDF • IEEE FORMAT

Full academic documentation: methodology, findings, and strategic recommendations.

03

Presentation Slides

.PPTX

Full capstone showcase slide deck built for stakeholder presentations.

04

Data Pipeline

.PBIX + CODE

Reproducible Power Query transformation, fully documented end to end.

1. Diagnose bottlenecks

2. Identify where value is lost

3. Quantify highest-impact interventions

Stage 01 — Multi-Source Data Ingestion.

Data is acquired from four source categories, integrated through API connections and structured database queries, with integrity validation applied at the point of entry.

- 1 Operational facility records — collection volumes & timelines
- 2 Sorting facility logs — material categorization & accuracy rates
- 3 Processing stage records — throughput, downtime, recovery yields
- 4 Administrative datasets — workforce & logistics

PIPELINE STAGE

01 / 04

METHOD

API connections + structured database queries

VALIDATION

Integrity validation at point of entry

MULTI-SOURCE

REPRODUCIBLE

Stage 02 — Cleaning & Feature Engineering.

Performed entirely in Microsoft Power Query (M Language) — transforming raw, inconsistent source records into analysis-ready fields.

OPERATIONS PERFORMED

- ✓ Duplicate removal
- ✓ Categorical label standardization
- ✓ Median imputation for missing values
- ✓ UTC normalization of timestamps

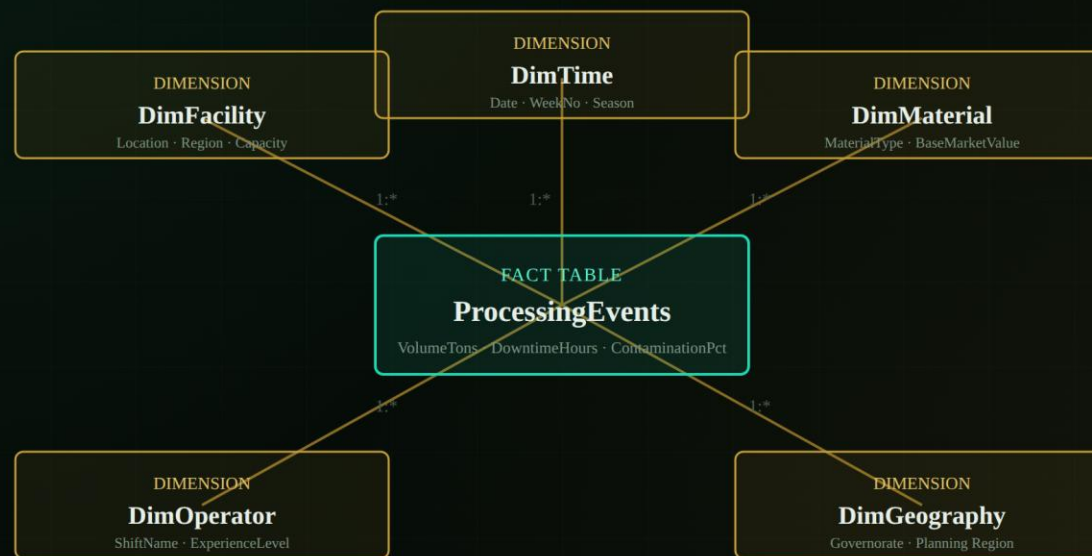
DERIVED FEATURES CREATED

- Collection efficiency ratios
- Sorting accuracy percentages
- Downtime duration aggregates
- SOURCE FILE
egypt_waste_observational_data_2021_to_2025.csv
COLUMNS: 25 ENCODING: UTF-8



Stage 03 — Dimensional Architecture: the **Star Schema**.

One central fact table — **ProcessingEvents** — surrounded by four dimension tables, giving every processed ton a time, a facility, a material, and an operator.



Stage 04 — Interactive Dashboard Architecture.

Built in Microsoft Power BI Desktop using a data-storytelling methodology across three dashboard pages, with cross-page synchronized slicers and drill-through to granular incident records.

PAGE 1

Executive Summary

AUDIENCE: SENIOR LEADERSHIP

Aggregate KPI cards with trend sparklines for collection efficiency, sorting accuracy, material recovery rate, and total downtime. Geographic heat map of facility-level performance variance.

PAGE 2

Downtime & Operational Analysis

AUDIENCE: OPERATIONS MANAGERS

Pareto chart of downtime root causes. Duration timelines by facility and cause category, with drill-through to individual incident records.

PAGE 3

Operator & Resource Performance

AUDIENCE: FACILITY SUPERVISORS

Scatter plots of operator tenure vs. throughput. Matrix visuals for cross-cohort comparison; performance broken down by shift and material type.

POWER BI DESKTOP

3 DASHBOARD PAGES

DAX MEASURES

CROSS-PAGE FILTERS

DRILL-THROUGH

Page 1 — Executive Summary

National Waste Performance, plotted live: recycling growth trajectory against the 2030 target, with the average monthly waste figure that anchors every governorate comparison in the deck.



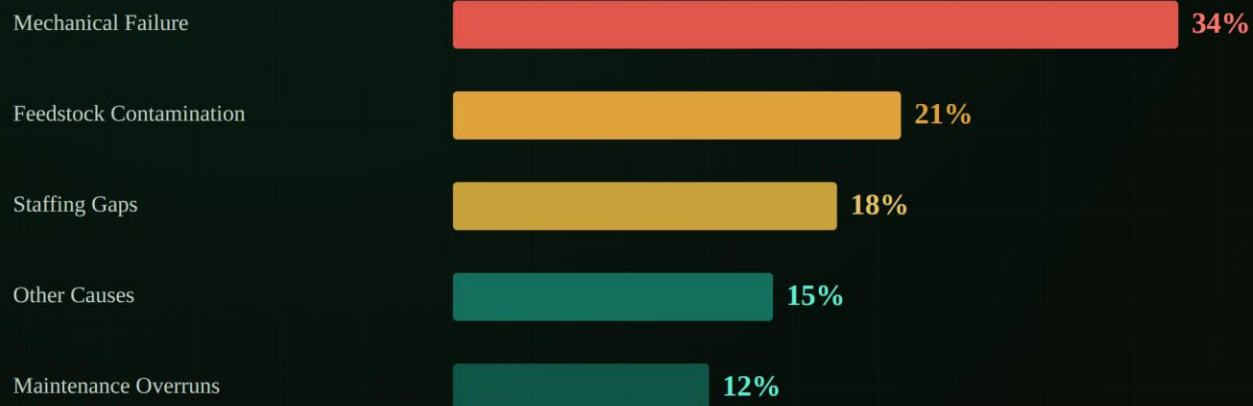
Recycling Growth Trajectory



Each KPI tile and chart element on the live dashboard carries hover/click drill-through hotspots, surfacing deeper analytical context — from a single national gauge down to a single governorate's monthly record.

Page 2 — Downtime & Operational Analysis

Operational Downtime by Root Cause — a Pareto view built directly from incident logs, showing exactly where lost processing hours originate.



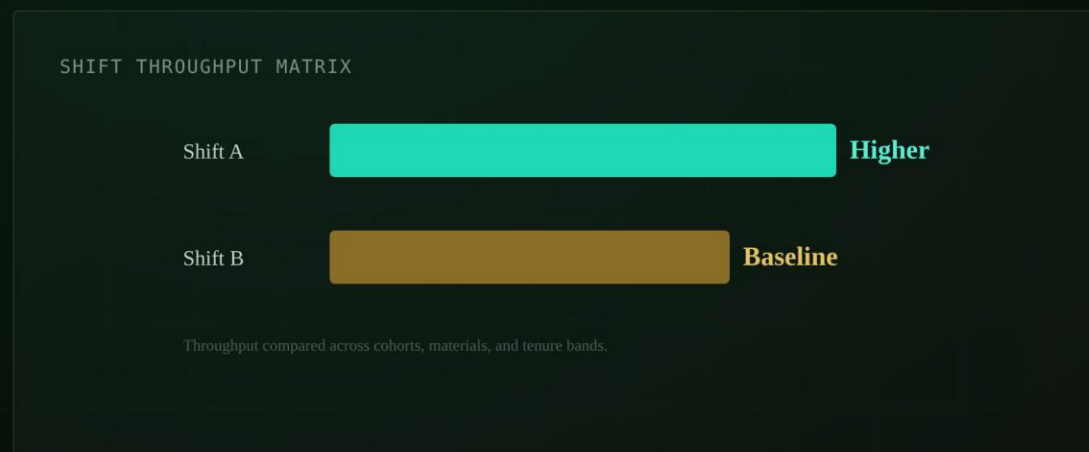
67%

of all downtime traces to just the **top 3 causes**: mechanical failure, contamination, and maintenance overruns.

Duration timelines break down further by **facility** and **cause category**, with drill-through to individual incident records for root-cause investigation.

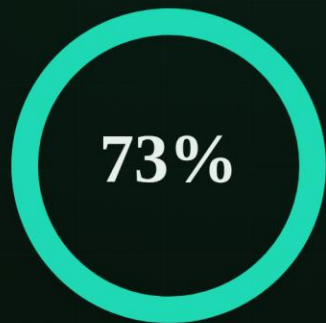
Page 3 — Operator & Resource Performance

Shift Throughput Matrix — comparing Shift A against Shift B — alongside a facility performance gap gauge used to flag statistical outliers.



Facility-level outlier flags are computed statistically — any facility more than 1.5 standard deviations from the portfolio mean recovery rate is marked for review, whether underperforming or a star performer worth studying.

Finding 01 — Collection Efficiency: 73% mean, high variance.



AVG COLLECTION EFFICIENCY

- Vehicle availability was the primary constraint — not workforce capacity.
- Performance troughs recur at the start of each calendar week (Monday) and during peak summer months.
- Root cause: fleet maintenance scheduling combined with heat-related vehicle failures.

IMPLICATION

Software-only route optimization can recover 8–12 efficiency points with zero capital investment — see Recommendation 03.

Finding 02 — Sorting Accuracy Drives **Everything** Downstream.

+1% Sorting Accuracy



× 2.3 multiplier



Every 1pt of sorting accuracy compounds into 2.3pts of recovered marketable material.

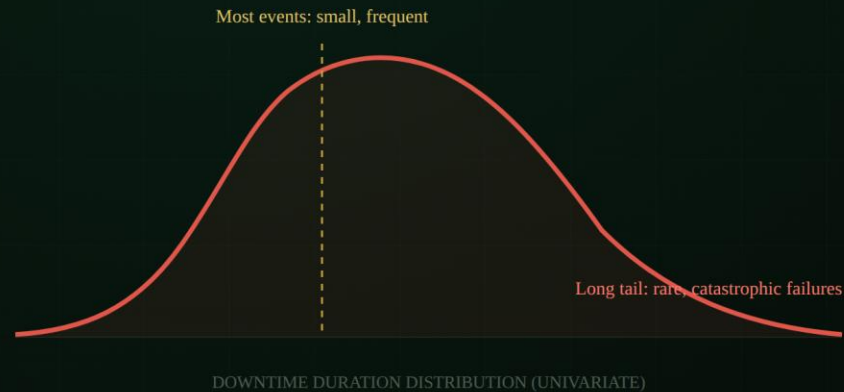
STATISTICAL RESULT

Bivariate analysis confirms a statistically significant **negative correlation** between sorting accuracy and downstream material contamination.

CONCLUSION

Front-end sorting quality is the *highest-ROI investment* in the entire system.

Finding 03 — Downtime is Highly Skewed, Not Evenly Spread.



- Univariate analysis reveals a **right-skewed distribution** in downtime duration.
- A small number of high-magnitude mechanical failure events cause disproportionately large losses.
- Routine minor delays are far more frequent — but far less impactful in aggregate.

CONCLUSION

Predictive maintenance is the primary intervention target.

Finding 04 — A 40%+ Gap Between Best and Average.

Top-Performing

Facilities

Top-Performing
Facilities

Portfolio
Average

Recovery Rate Baseline 40%+ gap

- Top-performing facilities achieve recovery rates **40%+ above** the portfolio mean.
- This gap is **not** explained by capital differences between facilities.
- It reflects operational practices, shift management, and sorting procedures — things people, not budgets, determine.

CONCLUSION

Best-practice dissemination is a zero-capital intervention with high potential.

Tab 1 — National Waste Performance

76.3k

AVG GOVERNORATE MSW (TONS/MO)

Cairo

HIGHEST MSW PRODUCER · 279.7K

55.8%

ORGANIC WASTE FRACTION

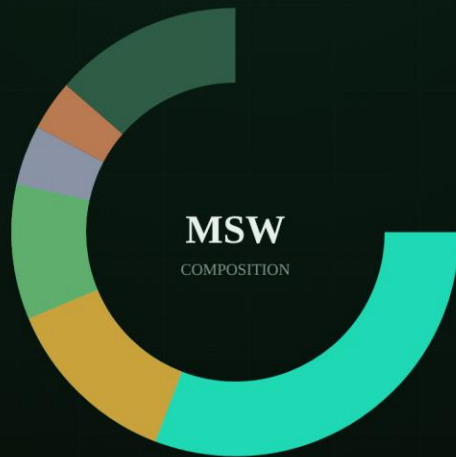
15.96%

MEAN RECYCLING RATE (BASELINE)

TOP 10 GOVERNORATES BY MONTHLY MSW GENERATION (TONS/MONTH)



Tab 1 cont. — Composition & the Governorate Map



Organic — 55.79%

Paper/Cardboard — 9.72%

Metal — 3.63%

Plastic — 12.89%

Glass — 4.33%

Other — 13.65%

INTERACTIVE MAP

A full SVG map of all 27 governorates is color-coded by recycling-rate intensity. Clicking any governorate — e.g. **Cairo: 279.8k tons/mo, 17.8% recycling, 66.3% coverage, 10.1M population** — opens a data card and an "AI Action Plan" button that triggers EcoBot for that region.

Tab 2 — Investment Opportunities

66.3%
CAIRO COVERAGE PEAK

36.0%
LOWEST COVERAGE · DELTA REGION

12.9%
PLASTIC FRACTION · HIGH-ROI SORTING

HIGH
INVESTMENT PRIORITY

WASTE COLLECTION COVERAGE BY REGION



DRY RECYCLABLE VOLUME

Cairo + Giza generate **over 500k tons** of waste combined monthly — the priority locations for modern Material Recovery Facilities (MRFs).

INFRASTRUCTURE GAP

Delta and Upper Egypt have huge generation volumes but only **~36% collection coverage**. Investing in municipal collection networks here yields the largest social and environmental returns.

Tab 3 — Research & Analytics

Strong

COVERAGE-RECYCLING LINK

\$3.7k

AVG GDP PER CAPITA

1,620

DATA POINTS ANALYZED

Star

DATA SCHEMA MODEL

RECYCLING RATE VS. WASTE COLLECTION COVERAGE



THE COLLECTION TRAP

Regions with low collection coverage face depressed formal recycling rates — because raw materials never enter formal processing facilities.

INCOME EFFECTS

Higher GDP-per-capita areas yield significantly higher plastic and packaging fractions, shifting processing needs from composting toward sorting.

5-YEAR TIME SERIES · ALL 27 GOVERNORATES · 1,620 DATA POINTS VIA POWER QUERY

HIGH IMPACT REC 01

Sorting Infrastructure Prioritization

DATA BASIS

The 2.3× sorting multiplier, plus a strong negative correlation between sorting accuracy and downstream contamination.

ACTION

Invest in near-infrared (NIR) spectroscopy systems for automated material identification, plus quality-control sensors at sorting conveyors.

PRIORITY

Highest ROI in the entire system.

PROJECTED IMPACT

+15–20

percentage points toward the Vision 2030 target

HIGH IMPACT REC 02

Predictive Maintenance Deployment

DATA BASIS

Mechanical failure accounts for 34% of all downtime; the right-skewed distribution shows a few catastrophic events cause the majority of losses.

ACTION

Deploy condition-monitoring sensor networks (vibration, temperature, pressure) on critical machinery, plus ML-based predictive maintenance models trained on historical failure patterns.

INDUSTRY BENCHMARK

Predictive maintenance reduces unplanned downtime by 30–50% across comparable industrial settings.

PROJECTED IMPACT

–30 to 50%

reduction in mechanical downtime

+10%+

recovered processing capacity

MEDIUM IMPACT REC 03

Route Optimization Algorithms

DATA BASIS

73% collection efficiency, with pronounced troughs on Monday mornings and through summer months — consistent with fleet maintenance gaps and heat-related vehicle degradation.

ACTION

Apply Vehicle Routing Problem (VRP) algorithms to collection scheduling.

CAPITAL REQUIRED

Zero — a software-only intervention.

PROJECTED IMPACT

73% → 85%+
TODAY PROJECTED

8–12 percentage point recovery in collection efficiency

MEDIUM IMPACT REC 04

Best-Practice Dissemination Program

DATA BASIS

A 40%+ gap between best and average facilities — not explained by capital differences (see Finding 04).

ACTION

A structured knowledge-transfer program leveraging existing human capital from top-performing facilities to average ones.

CAPITAL REQUIRED

Minimal — primarily a management and training initiative.

PROJECTED IMPACT

50%+

narrowing of the best-to-average performance gap

Data Governance Framework

DATA BASIS

Initial data quality review revealed non-standardized categorical definitions, varying timestamp formats, and non-random missing data patterns.

ACTION

Establish uniform field definitions, mandatory entry validation, quarterly data quality audits, and a central data stewardship role.

PROJECTED IMPACT

*Enables and amplifies
all other interventions*

Foundational prerequisite for everything else

COMBINED IMPACT OF ALL FIVE INTERVENTIONS

+15–20 points

in effective material recovery rate

Sufficient to bridge the gap to the 2030 target — subject to implementation fidelity and infrastructure readiness.

$$\begin{array}{ccccc} 37\% & + & 15\text{--}20\text{pts} & = & 52\text{--}60\% \\ \text{2024 BASELINE} & & \text{COMBINED LIFT} & & \text{PROJECTED RANGE} \end{array}$$

An Interactive Sandbox for Closing the Gap

Found on the live site's Recommendations page: three sliders let stakeholders model the combined effect of REC 01–03 in real time, against the 60% Vision 2030 target.

Fleet Routing Optimization (REC 03)

73% → 90%

Per 1% increase: +0.25% recycling rate · +1.8M EGP ROI · +25,000 tons diverted

Sorting Accuracy & Technology (REC 01)

88% → 99%

Per 1% increase: +1.2% recycling rate (2.3× multiplier) · +8.5M EGP ROI · +120,000 tons diverted

Equipment Maintenance Uptime (REC 02)

67% → 90%

Per 1% increase: +0.5% recycling rate · +3.2M EGP ROI · +50,000 tons diverted

PROJECTED EFFECTIVE RECYCLING RATE

37.0% → 60%+

"Target Met! 🎯" badge at ≥60%

ESTIMATED ANNUAL RETURN

M EGP

Combined cost savings & material sales

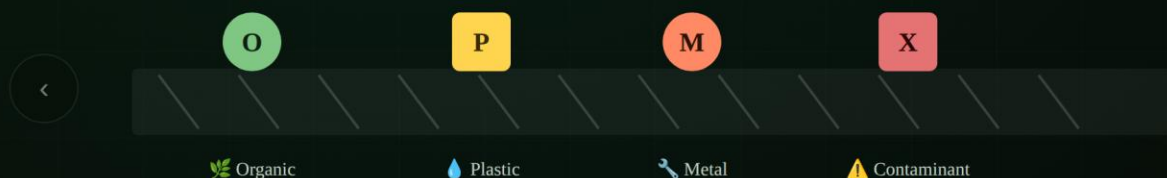
LANDFILL DIVERSION GAIN

Tons / Year

Additional waste diverted annually

Conveyor Belt **Sorting Game**

An interactive, Canvas-based educational game that makes the 2.3× sorting-accuracy finding tangible — click incoming materials to route them to the correct bin before they fall into the landfill.



45s

SHIFT DURATION

+10

PTS PER CORRECT SORT

-15

PTS PER CONTAMINANT CLICK

HUD tracks **Score**, **Accuracy %** (correct sorts ÷ total non-contaminant items), and **Tons Sorted** (+0.1t per correct sort).

END-OF-SHIFT RESULTS

Calculates a **yield multiplier** (accuracy × 2.3) — directly demonstrating Finding 02 from real operational data.

HTML5 CANVAS (800×320PX) · GameItem / GameParticle / GameFloatText classes

EcoBot — Present on Every Page

A floating AI chat assistant powered by the Google Gemini API, with a structured system prompt and a full offline keyword-matching fallback.

CONFIGURATION

API Provider	Google Gemini (gemini-pro)
Endpoint	/api/chat
Temperature	0.7
Max Output Tokens	500
Timeout	15s → offline fallback

SYSTEM PROMPT RULES

1. Conversational but informative, concise (2–4 sentences)
2. Team questions → list all 5 members with their cities
3. Strictly refuses off-topic questions, redirects to the project
4. Never fabricates data — only the facts in its system context

SUGGESTED QUESTIONS

"What is EcoMetrics Egypt?"

"What were the key findings?"

"Who are the team members?"

"How does the BI pipeline work?"

"What is Egypt's Vision 2030 target?"

On the Analysis page, the "AI Action Plan" button calls `window.triggerBotMessage()` to pre-fill the chat with a selected governorate's stats and asks EcoBot for a tailored recycling improvement plan.

Built on a **Zero-Dependency** Frontend, Real Analytics Tools.

FRONTEND

HTML5 (9 pages)
Vanilla CSS (shared.css, 69KB)
Vanilla JS (shared.js, 578 lines)
CSS custom properties for theming
Google Fonts: Space Grotesk, Inter, JetBrains Mono

UI LIBRARIES (CDN)

Chart.js 4.4.0 — all charts
Lucide Icons — all SVG icons
tsParticles Slim 2.0.6 — hero particle field

BACKEND & AI

Google Gemini API (gemini-pro)
Serverless function (api/chat.js)
GEMINI_API_KEY environment variable

DATA & ANALYTICS

Microsoft Power BI Desktop
Power Query (M Language)
DAX (Data Analysis Expressions)
Star Schema modeling
CSV source data (2021–2025)

DESIGN TOKENS



#060B08 · #00E5CC · #C9A84C · #FF4444 · #F59E0B

27 Governorates.

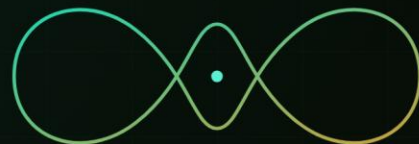
Five Years of **Observation.**

A sample of the underlying dataset (2021–2025), exported per governorate with population, waste, recycling, and coverage metrics.

GOVERNORATE	AVG MSW (TONS/MO)	AVG RECYCLING RATE	AVG COVERAGE	POPULATION
Cairo	279,780	15.96%	82.01%	10,636,817
Giza	255,334	15.94%	80.37%	9,707,387
Alexandria	158,604	15.87%	81.44%	5,679,854
Sharqia	117,707	16.04%	37.00%	8,055,066
Dakahlia	105,634	16.01%	36.91%	7,228,905
Beheira	101,107	15.96%	35.79%	6,919,095
+ 21 more governorates...	—	—	—	—

CSV EXPORT FORMAT (PER GOVERNORATE, 12 MONTHS):

Date, Governorate, Population, Total_MSW_Generated_Tons_Month, Recycling_Rate_Percentage, Waste_Collection_Coverage_Percentage, Organic/Plastic/Paper/Glass/Metal/Other_Percentage



Turning Waste Into Data. *Data Into Action.*

Data analytics transforms waste management from a reactive environmental concern into a proactive, evidence-based strategic asset — in support of Egypt Vision 2030.

`github.com/NotTheNix/Waste-Management-Recycling` · DEPI Data Analytics Capstone · 2024–2025

Supervised by Eng. Mahmoud Seraj · Computik Learning Solutions